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Cut it so it locks: Japanese joinery and the world's oldest wooden building

The free guide to the reel on Horyu-ji and Japanese carpentry: what makes the Kondo the oldest wooden building, how a notched joint locks two beams, what the five-storey pagoda tells us, and a paper-and-cardboard joint to cut yourself.



The oldest wooden building

The Kondo (main hall) of Horyu-ji in Nara is widely recognised as the world's oldest wooden building. It was rebuilt at least 1,300 years ago and measures 18.5 m by 15.2 m on two storeys. After a fire in 1949, about 15 to 20 percent of the original 7th-century material remains.

Beside it stands a five-storey pagoda, 32.45 m (122 feet) tall. Tree-ring analysis in 2001 showed that its central pillar (shinbashira) was felled in 594.

How Japanese carpentry works

Japanese carpentry was developed in the 12th century in the form we know today, influenced by Chinese architecture but evolving independently because of differences in earthquakes and climate. It is characterised by wooden joints and almost no nails.

Miyadaiku, the carpenters who build shrines and temples, are known for elaborate wooden joints, and their buildings are frequently among the world's longest surviving wooden structures.

Why a notch locks

Two beams butted end to end separate under a pull, because nothing resists it. Cut a stepped notch in both ends and they overlap: when the beams are pulled apart, the steps bear against each other. A small wooden key driven into the joint stops the beams sliding back.

This is the idea in the reel's diagram: two plain beams separate, the notched ones interlock and hold under the same pull.

Built examples

Horyu-ji Kondo, Nara	Widely recognised as the world's oldest wooden building.
Horyu-ji five-storey pagoda	32.45 m tall; its central pillar was felled in 594 (dendrochronology, 2001).
To-ji pagoda, Kyoto	The tallest wooden pagoda, five storeys and 54 m.

Try it yourself: cut a locking joint

1. Cut two strips of cardboard, each 20 cm by 4 cm.
2. At one end of each strip, cut a step: remove a 4 cm by 2 cm rectangle from the top of one strip and from the bottom of the other, so the steps interlock.
3. Slide the two ends together and pull the strips apart: the steps catch.
4. Slide a small card key through a slot cut across both overlapping ends, then pull again.
5. Try the same with two strips butted end to end and tape-free: compare how much pull each holds.

The rule to remember

Shape the parts so the load locks them; don't add a fastener to fight the load.

Default answer	Add nails or screws.
Form-finder answer	Cut the parts so that pulling them tightens the joint.
Check	Ask: which load would separate these parts, and can I make the shape resist exactly that?

Go further

Horyu-ji (Wikipedia)
https://en.wikipedia.org/wiki/H%C5%8Dry%C5%AB-ji

Japanese carpentry (Wikipedia)
https://en.wikipedia.org/wiki/Japanese_carpentry

Japanese pagoda (Wikipedia)
https://en.wikipedia.org/wiki/Japanese_pagoda

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